

# Does a Music Transformer Know What Key It Is In?

## Causal Interventions on an Emergent Tonal World Model

Anonymous ICASSP 2027 submission

**Abstract**—Music generation models produce convincingly tonal music. Whether they do so by chaining surface statistics or by maintaining an internal representation of musical key that generation *causally uses* has not been tested. We test it directly: a decoder-only Transformer trained on symbolic music from a functional-harmony grammar whose vocabulary contains absolute pitch and time only—no key, chord, or degree tokens—under a pre-registered protocol with frozen decision rules. Our central result is causal: replacing the key component of the residual stream at a single middle layer makes the model continue in the injected key at  $5.6\times$  the rate of rank- and norm-matched random edits (guarded strict TKR 0.378 vs. 0.068; significant for 12/12 target keys after Holm correction, replicated across seeds), while staying inside a musicality budget frozen before any edit ran—reaching 58% of the behavioral ceiling obtained by transposing the prompt itself. A capacity sweep sharpens what this means. Next-token accuracy is flat from 1.6M to 85M parameters, yet at 1.6M the key is not decodable beyond the input surface at all, and the same edit moves the key *more* often than at 25M while destroying the music in 90% of cases. Capacity does not buy the key; it buys a key that can be moved without breaking everything else. Finally, a matched control settles what generalizes: on real Bach chorales our synthetic-trained model does *not* beat the pitch surface, while a public model of *identical architecture* trained on real music does (margin +0.198, CI excluding zero)—with the training data as the single free variable. Editing a music Transformer’s internal key state changes what key it composes in: tonality is tracked as a causal latent variable, not just surface statistics.

**Index Terms**—music generation, world models, interpretability, causal intervention, tonality

### I. INTRODUCTION

Systems that compose music now write convincingly tonal pieces: their outputs settle into a home key, build tension away from it, and return. For listeners this is much of what makes generated music sound like music; for creators it determines whether such systems can be steered.

How they achieve this is largely unexamined. Existing analyses report that musical attributes can be *read out* of a model’s activations by trained classifiers [7]. Reading, however, is not using: a signal can be decodable from a system’s state without the system consulting it when it acts—a gap documented repeatedly in interpretability research [11]. No study has tested whether a music model

uses an internal key representation: whether intervening on it, while leaving the input untouched, changes the key of the music the model goes on to generate.

Sequence models can build causal models of the latent state behind their data. A model trained only to predict Othello moves tracks the board those moves imply, and editing that internal board changes which moves it treats as legal [1], [2]. Analogous causal evidence exists for chess, mazes, and RL environments [4], [5], [6], but not for music.

We close that gap. We choose key—tonic and mode—as the state variable for three reasons. First, unlike any candidate in the game domains, key is defined *externally* to the model, by centuries of theory and by psychological measurement [10], so the probed state is not an artifact of our own labeling. Second, among musical state candidates it is the most demanding: metric position is recoverable from time tokens, and harmonic function presupposes a key, whereas key must be *computed* by integrating pitch evidence over time. Third, key admits a natural analogue of Othello’s legal-move rate—diatonic scale membership—so causal success is measurable without a learned judge.

Our contributions:

- 1) A controlled testbed for world-model analysis in music: a leak-free tokenizer, a functional-harmony corpus with exact per-token key labels, and a pre-registered protocol, released as a benchmark (Sec. III).
- 2) Existence and geometry: key is decodable beyond strong input-side baselines in all six trained models, the twelve tonic directions self-organize into a circle of fifths, and—a pre-registered negative result—transposition augmentation does not make the representation a cyclic group (Sec. IV-A).
- 3) Causality: single-layer subspace edits redirect the composed key for 12/12 targets within a frozen musicality budget, and a capacity sweep shows that what capacity supplies is not the key but the ability to move it surgically—dissociating three things the term “world model” usually conflates (Secs. IV-C–IV-E).
- 4) A matched-architecture control on real music: with corpus, protocol, and decision rule held fixed, a public real-trained model clears the input-surface baseline where our synthetic-trained model of the

TABLE I  
TERMINOLOGY. DECISION RULES (DR-\*) WERE FROZEN  
PRE-EXPERIMENT.

|                               |                                                                      |
|-------------------------------|----------------------------------------------------------------------|
| $\kappa(t)$                   | Key state: tonic ( $\mathbb{Z}_{12}$ ) $\times$ mode at token $t$    |
| $\kappa^*$                    | Donor key injected by an edit                                        |
| D-SYN                         | Synthetic corpus with per-token ground-truth $\kappa(t)$             |
| M-CTRL                        | Studied model                                                        |
| M-REF                         | Frozen reference model; used only for the quality guard              |
| Edit                          | $h \leftarrow h - P_V h + P_V \mu_{\kappa^*}$ in the residual stream |
| TKR                           | Rate at which the continuation’s estimated key = $\kappa^*$          |
| IKR <sub><math>x</math></sub> | Fraction of continuation pitches diatonic to key $x$                 |
| $\delta_{\text{PPL}}$         | M-REF perplexity budget, frozen pre-intervention                     |

same shape does not, isolating the training data as the variable that decides whether a key state is worth more than the notes themselves (Sec. IV-F).

## II. RELATION TO PRIOR WORK

Prior work divides along two axes: domain and evidence level. Across domains, world-model analyses have progressed from readability to causal intervention in board games [1], [2], [3], [4], mazes [5], and RL environments [6]. In music, evidence has stayed at the readability level: probing studies decode theory concepts from generation models [7], and steering studies control attributes without testing a state variable [8]. We occupy the empty cell—causal, state-level evidence in music—for a state whose transitions (modulations) themselves follow a grammar, unlike the game domains.

## III. METHOD

### A. Problem statement and terminology

The input is a symbolic music sequence; the latent state is the running key  $\kappa(t)$ ; the question is whether the generator’s continuation causally depends on its internal estimate of  $\kappa$ . Table I fixes the terms. The pre-registration (supplementary) specifies every decision rule, threshold, and control, all frozen before any experiment ran; deviations are logged and negative results reported.

### B. Data and models

To make ground truth exact rather than estimated, all training data are synthetic. **D-SYN**. A functional-harmony state machine ( $T \rightarrow S \rightarrow D \rightarrow T$  with substitutions, SATB voicing, optional diatonic melody) generates 200k/10k/10k train/val/test pieces of 278–508 tokens with 0–3 modulations each (pivot-chord, direct, or sequential; fifths-distance stratified), giving every token a ground-truth key label. **Tokenizer**. To prevent state leakage the vocabulary (124 types) contains only BAR, POS, absolute-MIDI PITCH, and DUR tokens; a unit test enforces the absence of key/chord/degree tokens. **Models**. M-CTRL is

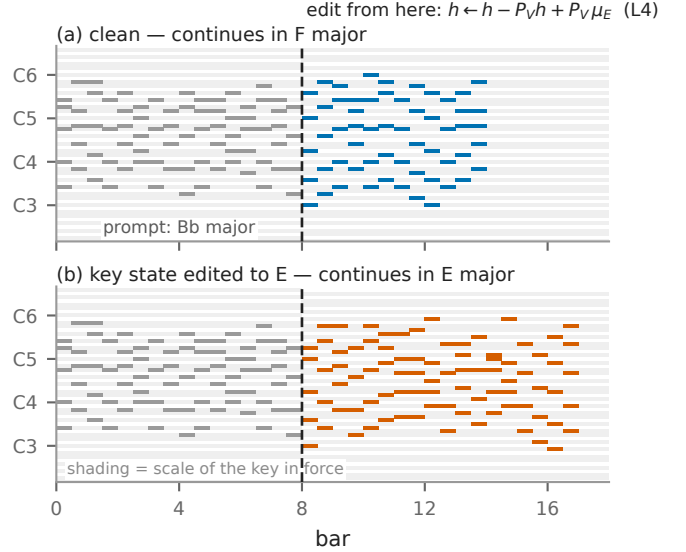


Fig. 1. The claim in one example (real model output, fixed sampling seed [s0]). The same 8-bar Bb-major prompt, continued (a) clean and (b) with the key component of layer-4 activations replaced by the E-major class mean from bar 9 on. **The input does not change**; the continuation’s estimated key follows the injected internal state. Shading marks the scale of the key in force.

a GPT-2 style decoder (8 layers, 8 heads,  $d = 512$ , context 512,  $\sim 25\text{M}$  parameters), trained 12k steps under two regimes—with (R-Aug) and without (R-NoAug) uniform 12-fold transposition augmentation— $\times 3$  seeds. M-REF shares the architecture but uses a disjoint seed *and* data split, and serves only as the musicality guard, avoiding circular evaluation. All models pass a pre-frozen quality gate (validation next-token top-1 0.878–0.881 vs. a 0.434 constant-predictor rate; unconditional in-key ratio 0.964–0.977 vs. a 0.608 threshold).

### C. Phase A: probing and geometry

To separate internal computation from input transcription, every probe result is reported against three control families. Linear (multinomial logistic) probes map residual-stream activations  $h_\ell(t)$  to  $\kappa(t)$  (24 classes) with sequence-level splits. **C1** permutes key identities per sequence—a control task in the sense of [11]; **C2** is an untrained model; **C3** are input-side baselines at the same positions (pitch-class histogram logistic regression, and Krumhansl–Schmuckler (KS) template matching; windows  $W \in \{8, 16, 32, 64\}$ ). DR-H1: the probe supports H1 iff its selectivity-corrected  $F_1$  exceeds the best C3 baseline with a sequence-level BCa 95% CI excluding zero in some layer. Ambiguity is stratified by KS posterior entropy ( $W=16$ ). For geometry, orthogonal Procrustes maps  $R_k$  are fit between activations of pieces and their  $k$ -semitone transpositions, and a cyclicity error tests group structure:

$$\varepsilon_{\text{cyc}} = \frac{1}{11} \sum_{k=2}^{12} \frac{\|R_k - R_1^k\|_F}{\|R_k\|_F}, \quad (1)$$

with DR-H2b:  $\varepsilon_{\text{cyc}}(\text{R-Aug}) < \varepsilon_{\text{cyc}}(\text{R-NoAug})$  (one-sided Wilcoxon).

#### D. Phase B: causal intervention

To test *use* rather than *presence*, we edit the state and score the behavior. From bar 9 of a key-stable 8-bar prompt, a sustained edit replaces the key component at one layer,  $h \leftarrow h - P_V h + P_V \mu_{\kappa^*}$ , with  $V$  an orthonormal key subspace and  $\mu_{\kappa^*}$  the class-conditional mean of the donor key. Subspaces: **V-PROBE** (probe row space, rank 24), **V-MEAN** (class-mean span), **V-DAS** (interchange-trained, ranks 8/24). Controls: **K1** rank/norm-matched random subspace; **K2** sham edit, required bit-identical to the clean run (passed, 100/100 prompts); **K3** a different layer’s subspace; **K4** behavioral reference, transposing the prompt itself. Continuations (16 bars; 12 major targets  $\times$  100 prompts  $\times$  8 layers; sampling noise paired with a clean twin) are scored by KS key estimation (TKR, strict and closely-related-tolerant), IKR, and a grammar guard. The budget  $\delta_{\text{PPL}} = 0.613$  (90th percentile of the M-REF perplexity rise across 7,989 natural modulations in validation data) was fixed *before* any edit ran; edits exceeding it do not count as successes. DR-H3 requires  $\text{TKR}(\text{edit}) > \text{TKR}(\text{K1})$  with Holm-corrected Wilcoxon  $p < .05$  in  $\geq 8/12$  targets at the best in-budget condition.

### IV. RESULTS

Intervention numbers are from the R-Aug seed-0 sweep [s0] unless noted; probing holds for all six M-CTRL models, and the seed-1 sweep replicates every Phase-B verdict (Sec. IV-E).

#### A. H1: the key is decodable beyond the input surface

The key is *computed* by the model, not transcribed from the input: the corrected probe–baseline margin excludes zero in layers 1–7 of every trained model (peak +0.105) and fails only at L0 (−0.20). Table II gives the numbers at the layer the edit acts on. The internal state is most valuable where the surface is least informative: in the top ambiguity quartile probe accuracy is 0.936 vs. 0.795 for the best baseline.

#### B. H2: fifths geometry; equivariant but not a group

The representation carries music-theoretic structure, but not the strictest form we pre-registered. Projected to two principal components, the twelve major-tonic directions order as C–G–D–A–E–B–F♯... (Fig. 2): circle-of-fifths neighbour agreement 1.00 vs. 0.00 for the chromatic ordering, at every probed layer (exploratory). Transposition acts as a generalizing linear map: Procrustes  $R_k$  fit on half the sequences aligns held-out activations with 0.05–0.11 relative error. The family is not cyclic, however:  $\varepsilon_{\text{cyc}} \approx 0.98$  vs.  $\approx 1.41$  for random orthogonal maps [Eq. (1)]. Our pre-registered comparison is negative: augmentation does not improve cyclicity (3 seeds per regime; R-Aug 0.983 vs. R-NoAug 0.981; one-sided  $p = 0.83$ ), so DR-H2b is *not* supported. The representation

TABLE II

KEY DECODING AT L4 (R-AUG SEED 0), 24-CLASS MACRO- $F_1$  ON HELD-OUT SEQUENCES; DR-H1 SUPPORTED IN 6/6 MODELS. NOTE THE TWO C1 ROWS MEASURE DIFFERENT THINGS: A CONTROL PROBE TRAINED ON PERMUTED LABELS BARELY LEARNS THEM (0.040, NEAR THE 1/24 CHANCE RATE—THE CONTROL TASK IS GENUINELY UNLEARNABLE), AND PREDICTS THE *TRUE* KEY AT 0.032—THE SELECTIVITY FLOOR THE MARGIN SUBTRACTS. THE MARGIN RECONCILES EXACTLY:  $0.927 - 0.032 - 0.790 = 0.105$ .

| Method / baseline                                     | macro- $F_1$ |
|-------------------------------------------------------|--------------|
| Linear probe on $h_4(t)$                              | 0.927        |
| MLP probe (secondary)                                 | 0.932        |
| Best input baseline (C3: PC-hist LR, $W=64$ )         | 0.790        |
| Untrained model (C2)                                  | 0.243        |
| C1 control probe, on its own permuted labels          | 0.040        |
| C1 control probe, on the true key (selectivity floor) | 0.032        |
| Corrected margin, BCa 95% CI [0.091, 0.114]           | +0.105       |

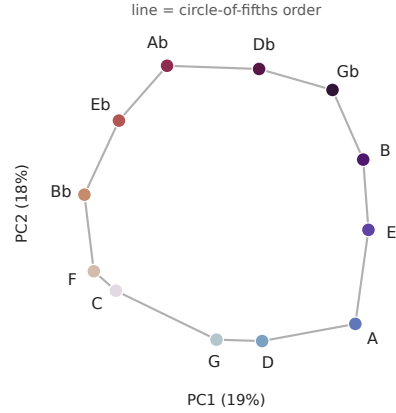


Fig. 2. The twelve major-tonic probe directions (L4, first two PCs) self-organize into a circle of fifths. No key, chord, or degree token exists in the training vocabulary.

is transposition-equivariant in effect but not organized as a cyclic group, and the key diversity already present in the data appears to suffice.

#### C. H3: editing the key state changes the composed key

Editing the internal key redirects generation. At the best in-budget condition (V-PROBE, L4),  $\text{TKR}(\text{edit}) - \text{TKR}(\text{K1}) > 0$  holds for 12/12 targets (Holm-corrected  $p < 10^{-4}$  each; rank-biserial  $r = 0.64$ –1.00) against a pre-registered bar of  $\geq 8/12$ : DR-H3 is supported. Fig. 3 gives all conditions. Pitch content triangulates the same conclusion without the estimator:  $\text{IKR}_{\kappa^*} = 0.937$  vs.  $\text{IKR}_{\kappa_{\text{src}}} = 0.646$  after editing. Success falls gently with target distance on the circle of fifths (0.395 at distance 1 to 0.280 at the tritone), while random edits collapse to zero once the identity target is excluded. The edit reaches 58% of the behavioral ceiling (K4), quantifying an *actionability gap*: the state is readable in a rank-24 subspace of one layer, yet a single-layer edit realizes only part of the behavioral effect. Two subsidiary findings: the

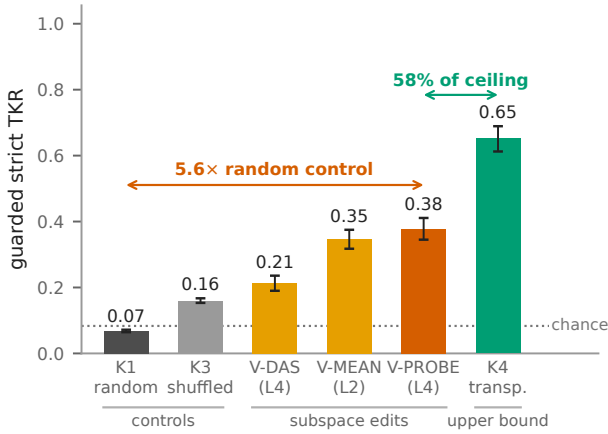


Fig. 3. Guarded strict TKR across conditions [s0]: the continuation lands exactly on the injected key *and* stays within the frozen musicality budget. Chance = 1/12. Bars: prompt-level bootstrap 95% CI.

optimized subspace (V-DAS) underperforms the probe’s readout directions (Sec. V), and V-MEAN edits destroy output quality at deep layers (guard pass  $\leq 5\%$  at L4–L5)—the guard is load-bearing, a point Sec. IV-E makes decisive.

#### D. H5: readability is diffuse; causal efficacy is sharp

The layer profile of the edit-minus-control effect is single-peaked (0.03, 0.04, 0.19, 0.26, **0.30**, 0.26, 0.23, 0.23 for L0–L7); the best-vs. -median bootstrap CI [0.038, 0.109] excludes zero, so DR-H5 is supported. The two profiles in Fig. 4 are *not* the same shape, and the difference is the finding. The probe saturates across L2–L6 (macro- $F_1$  0.920–0.929, all within 0.01 of its maximum), so the key is legible from most of the network; yet a single-layer edit moves the output only near L4, and barely at all at L0–L1. Where a state is legible is a weak guide to where it is used—a caution for interpretability work that localizes a computation by probing accuracy alone.

#### E. Replication, and what capacity actually buys

**Second seed.** An independently trained R-Aug seed-1 model reproduces every Phase-B verdict: DR-H3 supported with V-PROBE significant in 12/12 targets (TKR 0.23–0.41 vs. K1 0.02–0.10), and DR-H5 supported (CI [0.122, 0.181]). Its profile peaks at L2 rather than L4: the *localization* replicates, its exact depth does not, and we report both profiles rather than a pooled one.

**Capacity.** Training the same data and protocol at 1.6M (L2,  $d=128$ ), 6M (L4,  $d=256$ ), 25M (main line), and 85M (L12,  $d=768$ ) parameters dissociates three things the term “world model” usually conflates (Fig. 5). *Behavior* is flat: next-token top-1 is 0.868–0.879 across a  $50\times$  capacity range. *Readability* is not: at 1.6M the probe peaks at  $F_1$  0.63–0.66, *below* the input baseline (0.79), and DR-H1 is not supported in either seed—a behaviorally equivalent

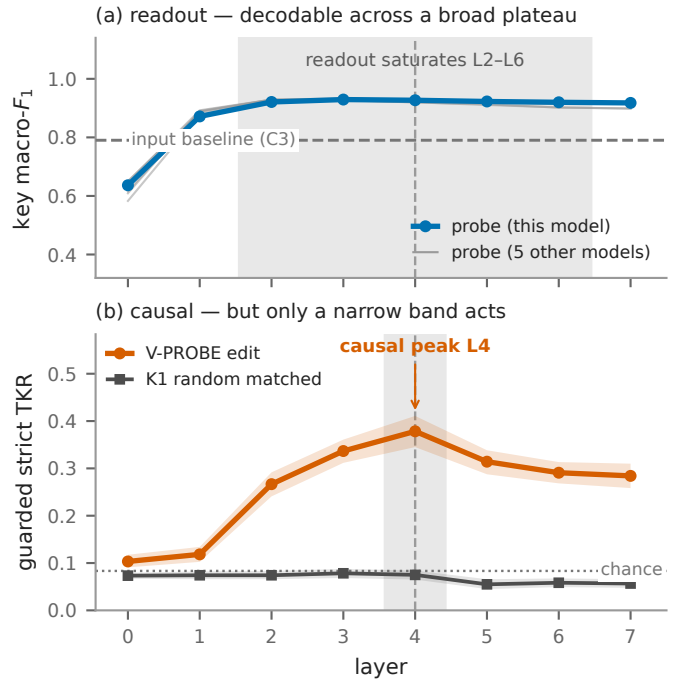


Fig. 4. (a) Probe macro- $F_1$  per layer, all six M-CTRL models (thin lines; R-Aug seed 0 highlighted): the key is legible across a broad plateau. (b) Causal effect [s0]: guarded strict TKR of the V-PROBE edit vs. the K1 matched random control (bands: prompt-level bootstrap 95% CI). Only a narrow band around L4 moves the output.

model that has no key state at all, which is the surface-statistics account made flesh, and which our pre-registered test correctly refuses to credit. *Manipulability* is a third thing again. At 1.6M the edit changes the key *more* often than at 25M in raw terms (0.55 vs. 0.43) and passes the musicality guard only 10% of the time (vs. 80% at 25M): with two layers, “editing the key subspace” is a blunt overwrite of the output, not the movement of a separable state. Reporting guarded TKR alone would hide this; reporting raw TKR alone would credit a surface model with a world model. **Capacity does not buy the key—it buys a key that can be moved without breaking everything else**, and the pre-registered guard is precisely the instrument that tells the two apart.

Manipulability does not simply keep improving, however. It peaks near 25M and then *falls*: the 85M model reaches guarded TKR 0.18, about half the 25M value. We suspected an artifact of our own making—each size model was edited at its probe- $F_1$  peak, a heuristic this paper itself refutes—so we swept all twelve layers of the 85M model. The hypothesis was wrong: its causal peak (L6, 0.206) is statistically indistinguishable from the layer we had already used (L3, 0.202). The decline is real. A natural reading, which we did not test and do not claim, is that greater depth leaves more downstream layers able to re-derive the key from the surviving context, so a single-layer edit is more easily repaired—the actionability gap

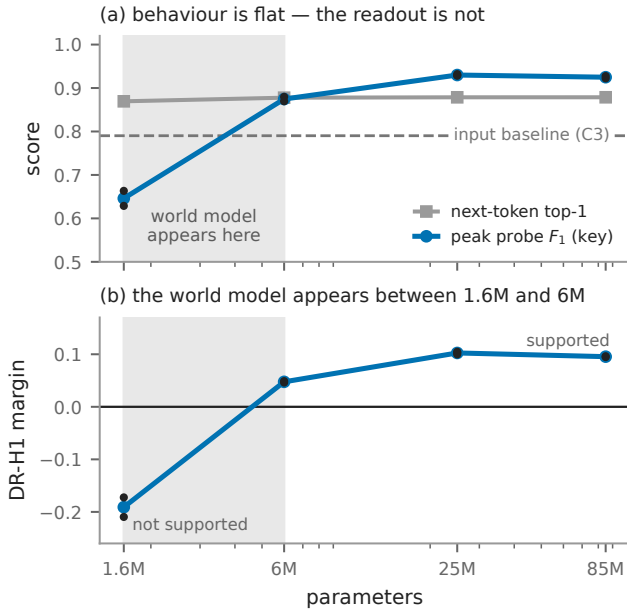


Fig. 5. Behavioral equivalence, representational dissociation. Across a  $50\times$  capacity range next-token accuracy is flat, but the key readout is not: below  $\sim 6M$  the probe cannot beat the input baseline and the pre-registered DR-H1 margin is negative. Dots: seeds.

widening with scale rather than closing. (Layer sweep: one seed.)

#### F. Real music, and a public model: what generalizes

Two questions remain for a synthetic-data study: does the key readout survive real music, and does any of this describe a model people actually use? We answer both with one matched comparison, on 300 Bach chorales whose *local* key at every note is given by published Roman-numeral analyses.<sup>1</sup>

**Our model does not clear the surface on real music.** Probed on the chorales, our 85M model reaches macro- $F_1$  0.523 against a selectivity floor of 0.037—the key subspace is real and transfers, it is not an artifact of our grammar—but the best input baseline (KS,  $W=64$ ) reaches 0.447, and the corrected margin, +0.034 (CI  $[-0.020, 0.100]$ ), does not exclude zero. DR-H1 is *not* supported out of distribution.

**A public model trained on real music does clear it.** The Anticipatory Music Transformer [9] (Apache-2.0; Lakh MIDI, MetaMIDI, FMA, and 450k commercial records) is, at its **small** size, *architecturally identical* to our 85M model: 12 layers,  $d=768$ . Its vocabulary is (arrival-time, duration, note) triples—leak-free in exactly the sense ours is. We probed it on the same chorales, at the same positions, with the same controls and the same decision rule; the two C3 baselines land at 0.447 and 0.446, confirming the setups are matched. It reaches macro- $F_1$  0.686 (L10)

<sup>1</sup>Scores: **bach-370-chorales** (CC BY-NC-SA); analyses: *When in Rome* (CC BY-SA). Neither is redistributed.

with a corrected margin of +0.198 (CI  $[0.133, 0.266]$ ): **DR-H1 supported**. With architecture, corpus, protocol, and decision rule held fixed, *the training data is the single free variable*, and it is what decides whether a key state beats the pitch surface. The layer profiles say the same thing from the other side: our synthetic model’s readout peaks early (L2) and decays to 0.17 by L11 on this out-of-distribution input, while the real-trained model’s dips mid-network and *builds* to its peak at L10.

Two cautions. The probe must be read where the model is about to *choose* a pitch (the token before a note); read at the note itself—after the pitch is committed, when the model is predicting the next onset—the same public model’s profile decays with depth and the key appears to vanish. Where one probes is not a free parameter. And this is a readability result: we have not intervened on the public model, so it inherits none of our causal claims.

## V. DISCUSSION AND LIMITATIONS

The three levels support the world-model reading *within this training condition*: a Transformer trained on notes alone maintains a mid-layer key variable with music-theoretic geometry, and generation consults it. The gap to the behavioral ceiling, the failure of edits at L0–L1, and the partial effect of layer-shuffled subspaces ( $K3 = 0.160$ , between K1 and the true edit) indicate a distributed, redundantly re-estimated state rather than a single canonical register—a contrast with Othello-GPT’s near-complete editability that we attribute to music’s continual re-grounding of tonality in the immediately preceding notes.

Four observations bound our claims. *V-DAS below V-PROBE*: the optimized subspace may simply be under-trained relative to its search space; we report it as observed and release its configuration, but do not claim optimization is inherently inferior. *Estimator*: the KS judge confuses fifths-neighbours, which strict/tolerant reporting brackets and IKR triangulates. *Capacity*: the sweep varies depth and width together, so it locates a threshold, not its cause. *External validity*: the causal results are for our own small, synthetic-trained models; Sec. IV-F shows the *readout* generalizes to a public real-trained model, but intervening on such a model is future work. Whether edited continuations modulate *musically* (pivot-like transitions vs. jumps) is measured in the journal version. All training data are synthetic; no artist recordings or copyrighted corpora are used for training, and the evaluation corpora are used under their stated licenses.

## VI. CONCLUSION

We gave pre-registered causal evidence that a music Transformer tracks musical key as an internal state variable that generation consults, and quantified how far a single-layer edit goes toward full behavioral control. The capacity sweep separates three properties usually bundled under “world model”—the state being present, being legible, and being separately manipulable—and shows they

emerge at different points. The testbed (exact state labels, leak-free vocabulary, frozen decision rules) is released as a benchmark. Because music offers state variables that theory defines and ears can verify, we expect it to be a useful substrate for asking what sequence models know; our next question is whether models also internalize how keys are *supposed* to change.

**Reproducibility.** Anonymized code, data generator, pre-registration, and an audio demo page (clean vs. edited continuations): <https://anonymous.4open.science/r/TBD>. **[TBD (run required): create before submission.]**

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